

LINGYU ZHU

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EDUCATION EXPERIENCE

City University of Hong Kong

Computer Science, PhD. Advisor: Shiqi Wang.

Hong Kong, China

2021.01 - 2024.12

The Hong Kong University of Science and Technology

Mechanical Engineering, Master. Advisor: Kai Tang.

Hong Kong, China

2018.09 - 2019.09

Wuhan University of Technology

Naval Architecture and Ocean Engineering, Bachelor. Advisor: Ziru Li.

Wuhan, China

2014.09 - 2018.06

VISITING AND WORKING ACTIVITIES

- **City University of Hong Kong.** Postdoctoral Research Fellow. Advisor: Shiqi Wang. Timeline: 2025.01 - 2025.11
- **Nanyang Technological University.** Visiting Student. Advisor: Weisi Lin. Timeline: 2024.10 - 2024.12
- **Peking University Institute of Information Technology.** Intern. Advisor: Zhao Wang. Timeline: 2023.06 - 2024.01
- **Peng Cheng Laboratory.** Intern. Advisor: Siwei Ma. Timeline: 2021.10 - 2022.10
- **City University of Hong Kong.** Research Assistant. Advisor: Shiqi Wang. Timeline: 2019.08 - 2021.01

RESEARCH INTERESTS

Learned Image/Video Coding for Human/Machine Vision, Large Multimodal Model, Generative Model, Quality Assessment, Computational Photography, etc.

PUBLICATIONS AND PRE-PRINT (CITATIONS $\geq$ 200)

- [1] **Lingyu Zhu**, Wenhan Yang, Xiangrui Zeng, Bolin Chen, Peilin Chen, Yung-Hui Li, Shiqi Wang: Leveraging Diffusion Knowledge for Generative Image Compression with Frequency-Aware via Fractal Band Learning, **International Conference on Computer Vision (ICCV)**, 2025. (Submitted).
- [2] **Lingyu Zhu**, Wenhan Yang, Danni Huang, Baoliang Chen, Hanwei Zhu, Bolin Chen, Long Xu, Zhao Wang and Shiqi Wang: Neural Feature Coding for Machine Vision in Low-Light Environments, **IEEE International Conference on Multimedia & Expo (ICME)**, 2025. (Submitted).
- [3] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Hanwei Zhu, Zhangkai Ni, Qi Mao and Shiqi Wang: Unrolled Decomposed Unpaired Learning for Controllable Low-Light Video Enhancement, **European Conference on Computer Vision (ECCV)**, 2024. (Poster).
- [4] **Lingyu Zhu**, Binzhe Li, Riyu Lu, Peilin Chen, Qi Mao, Zhao Wang, Wenhan Yang and Shiqi Wang: Learned Image Compression for Both Humans and Machines via Dynamic Adaptation, **International Conference on Image Processing (ICIP)**, 2024. (Oral).
- [5] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Hanwei Zhu, Xiandong Meng and Shiqi Wang: Temporally Consistent Enhancement of Low-light Videos via Spatial-Temporal Compatible Learning, **International Journal of Computer Vision (IJCV)**, 2024.
- [6] Baoliang Chen*, **Lingyu Zhu***, Hanwei Zhu, Wenhan Yang, Linqi Song and Shiqi Wang: Gap-Closing Matters: Perceptual Quality Evaluation and Optimization of Low-Light Image Enhancement, **IEEE Transactions on Multimedia (TMM)**, 2023. (*Co-first Author).
- [7] **Lingyu Zhu**, Wenhan Yang, Baoliang Chen, Fangbo Lu and Shiqi Wang: Enlightening Low-light Images with Dynamic Guidance for Context Enrichment, **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2022.
- [8] Baoliang Chen, Hanwei Zhu, **Lingyu Zhu**, Shanshe Wang, Jingshan Pan and Shiqi Wang: Debaised Mapping for Full-Reference Image Quality Assessment. **IEEE Transactions on Multimedia (TMM)**, 2025.
- [9] Bolin Chen, Shanzhi Yin, Zihan Zhang, Jie Chen, Ru-Ling Liao, **Lingyu Zhu** and Shiqi Wang, Yan Ye: Beyond GFVC: A Progressive Face Video Compression Framework with Adaptive Visual Tokens. **Data Compression Conference (DCC)**, 2025. (Oral)
- [10] Hanwei Zhu, Haoning Wu, Yixuan Li, Zicheng Zhang, Baoliang Chen, **Lingyu Zhu**, Yuming Fang, Guangtao Zhai, Weisi Lin and Shiqi Wang: Adaptive Image Quality Assessment via Teaching Large Multimodal Model to Compare. **Neural Information Processing Systems (NeurIPS)**, 2024. (Spotlight).
- [11] Hao Luo, Baoliang Chen, **Lingyu Zhu**, Peilin Chen and Shiqi Wang: RCNet: Deep Recurrent Collaborative Network for Multi-View Low-Light Image Enhancement. **IEEE Transactions on Multimedia (TMM)**, 2024.
- [12] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, Peilin Chen, Linqi Song and Shiqi Wang: Video Quality Assessment for Spatio-Temporal Resolution Adaptive Coding. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2024.

- [13] Baoliang Chen, Hanwei Zhu, **Lingyu Zhu**, Shiqi Wang and Sam Kwong: Deep Feature Statistics Mapping for Generalized Screen Content Image Quality Assessment. **IEEE Transactions on Image Processing (TIP)**, 2024. (Second Prize, Guangdong Computer Academy).
- [14] Riyu Lu, **Lingyu Zhu**, Baoliang Chen, Xiaopeng Fan and Shiqi Wang: Diffusion-based Bit-depth Expansion. **International Workshop on Multimedia Signal Processing (MMSP)**, 2024. (Oral).
- [15] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, and Shiqi Wang: Learning Spatiotemporal Interactions for User-generated Video Quality Assessment. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2022.
- [16] Baoliang Chen, **Lingyu Zhu**, Chenqi Kong, Hanwei Zhu, Shiqi Wang and Zhu Li: No-reference Image Quality Assessment by Hallucinating Pristine Features. **IEEE Transactions on Image Processing (TIP)**, 2022.
- [17] Liangwei Yu, Zhao Wang, Yan Ye, **Lingyu Zhu** and Shiqi Wang: A Soft-ranked Index Fusion Framework with Saliency Weighting for Image Quality Assessment. **The IEEE / CVF Computer Vision and Pattern Recognition Conference Workshop (CVPRW)**, 2022. (Second Place, 5th Workshop and Challenge on Learned Image Compression).
- [18] Baoliang Chen, **Lingyu Zhu**, Guo Li, Hongfei Fan and Shiqi Wang: Learning Generalized Spatial-temporal Deep Feature Representation for No-reference Video Quality Assessment. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2021.
- [19] Guo Li, Baoliang Chen, **Lingyu Zhu**, Qingwen He, Hongfei Fan and Shiqi Wang: PUGCQ: A Large Scale Dataset for Quality Assessment of Professional User-generated Content. **ACM MM**, 2021. (Poster).
- [20] Hanwei Zhu, Baoliang Chen, **Lingyu Zhu**, Shiqi Wang and Weisi Lin: DeepDC: Deep Distance Correlation as a Perceptual Image Quality Evaluator. **IEEE Transactions on Image Processing (TIP)**, 2025. (Major)
- [21] Bolin Chen, Hanwei Zhu, Shanzhi Yin, **Lingyu Zhu**, Jie Chen, Ru-Ling Liao, Shiqi Wang and Yan Ye: Pleno-Generation: A Scalable Generative Face Video Compression Framework with Bandwidth Intelligence. **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**, 2025. (Submitted).
- [22] Songchao Tan, **Lingyu Zhu**, Hanwei Zhu, Shiqi Wang, Xinfeng Zhang, Siwei Ma and Huimin Ma: Visual Feature Compression and Quality Evaluation for Machine Vision Analysis: A Review. **IEEE International Conference on Multimedia & Expo (ICME)**, 2025. (Submitted).
- [23] Hao Luo, **Lingyu Zhu**, Yudong Mao, Yixuan Li, Zhiwei Zhong and Shiqi Wang: Exploiting Long and Short Temporal Dependence for Low-Light Video Enhancement. **IEEE International Conference on Multimedia & Expo (ICME)**, 2025. (Submitted).
- [24] Siyi Pan, Danni Huang, Baoliang Chen, Hanwei Zhu, **Lingyu Zhu**, Xiangjie Sui and Shiqi Wang: Mitigating Perception Bias: A Training-Free Approach to Enhance LMM for Image Quality Assessment. **The IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)**, 2025. (Submitted)
- [25] Bolin Chen, Shanzhi Yin, Hanwei Zhu, **Lingyu Zhu**, Zihan Zhang, Jie Chen, Ru-Ling Liao, Shiqi Wang and Yan Ye: Compressing Human Body Video with Interactive Semantics: A Generative Approach. **International Conference on Image Processing (ICIP)**, 2025. (Submitted).
- [26] Hao Luo, Zhiwei Zhong, **Lingyu Zhu**, Yudong Mao and Shiqi Wang: MS-MoE: Multi-modal Structural Mixture of Experts Framework for Pan-Sharpening. **International Joint Conference on Neural Networks (IJCNN)**, 2025. (Submitted).
- [27] Xiangrui Zeng, **Lingyu Zhu**, Peilin Chen, Zhao Wang, Shiqi Wang and Sam Kwong: Examining the Holistic Capabilities of Large Vision Language Models in Low-Light Image Enhancement: Challenges and Beyond. **IEEE International Symposium on Circuits and Systems (ISCAS)**, 2025. (Submitted).
- [28] Qi Mao, Ruofan Wang, **Lingyu Zhu**, Chuanmin Jia, Shiqi Wang and Siwei Ma: DHVC: Disentangled Visual Representations for Extreme Human-Body Video Compression. **ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)**, 2025. (Submitted).

TEACHING ASSISTANT

- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2024.09 - 2024.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2024.01 - 2024.05
- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2023.09 - 2023.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2023.01 - 2023.05
- Semester A, CS3343, Software Engineering Practice, City University of Hong Kong. Timeline: 2022.09 - 2022.12
- Semester B, CS3342, Software Design, City University of Hong Kong. Timeline: 2022.01 - 2022.05
- Semester A, CS5481, Data Engineering, City University of Hong Kong. Timeline: 2021.09 - 2021.12
- Semester B, CS1102, Introduction to Computer Studies, City University of Hong Kong. Timeline: 2021.01 - 2021.05

MISCELLANEOUS

- Programming Skills: Python, Pytorch, Tensorflow, Matlab, C

HONORS AND AWARDS

- Second Prize, Guangdong Computer Academy. Timeline: 2024.12
- Outstanding Academic Performance Award. Timeline: 2024.08 - 2025.08
- Conference Grant. Timeline: 2024.09 - 2024.10
- Institutional Research Tuition Scholarship. Timeline: 2023.09 - 2024.09
- Workshop and Challenge on Learned Image Compression: 2nd Place in the Perceptual Metric Track. Timeline: 2022.06
- Research Activities Fund. Timeline: 2021.10 - 2022.04
- Research Activities Fund. Timeline: 2021.01 - 2024.12
- Wuhan University of Technology CSC Scholarship. Timeline: 2014.11 - 2015.11

PROFESSIONAL SERVICES

Journal Reviewer

- IEEE Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**)
- International Journal of Computer Vision (**IJCV**)
- IEEE Transactions on Image Processing (**TIP**)
- IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**)
- IEEE Transactions on Multimedia (**TMM**)
- IEEE Multimedia (**MM**)
- Journal of Visual Communication and Image Representation (**JVCI**)
- Computer Vision and Image Understanding (**CVIU**)
- Pattern Recognition (**PR**)
- Machine Vision and Applications
- Multimedia Systems
- Scientific Reports

Conference Reviewer

- ACM International Conference on Multimedia (**ACM MM**)
- IEEE / CVF Computer Vision and Pattern Recognition Conference (**CVPR**)
- European Conference on Computer Vision (**ECCV**)
- IEEE International Conference on Multimedia & Expo (**ICME**)
- IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**)
- International Joint Conference on Neural Networks (**IJCNN**)
- IEEE International Symposium on Circuits and Systems (**ISCAS**)